DIVINE GRACE COLLEGE OF ARTS, SCIENCE AND TECHNOLOGY (DGCAST) – GAROUA

GEOGRAPHY REVISION SHEET – FORM FOUR

First Sequence Evaluation, School Year 2026-2027 • Teacher: Mr KAMDEM Emmanuel

Lessons revised: Soils, erosion and conservation; ecosystems; savanna and hot desert biomes; situation and relief of Cameroon

A. THE SOIL: MEANING, COMPONENTS AND PROFILE

1. Define soil. What is it made of?

Answer: Soil is the loose top layer of the Earth's surface in which plants grow. It is formed by physical, chemical and biological processes and is made of **mineral particles** (about 45 %), **organic matter** or humus (about 5 %), and **water and air** (about half of the soil).

2. Name the horizons of a simple soil profile.

Answer: Topsoil (A): dark, rich in humus, most roots; **subsoil (B)**: little humus, minerals washed down from the topsoil; **regolith (C)**: weathered and broken rock; **bedrock (R)**: solid rock at the bottom. A deep soil rich in humus and minerals is fertile.

B. SOIL EROSION AND SOIL CONSERVATION

3. Define soil erosion and accelerated erosion. Name the forms of erosion.

Answer: Soil erosion is the removal of the topsoil by water, wind or people. **Accelerated erosion** is erosion made faster by human activities, while normal (geologic) erosion is slow and natural. Forms: sheet, rill, gully and wind erosion.

4. Give the causes and the consequences of soil erosion.

Answer: Physical causes: heavy rain, steep slopes, strong winds. Human causes: deforestation, overgrazing, bush fires, ploughing up and down slopes, over-cultivation. Consequences: the soil loses its fertility and harvests fall; gullies destroy land and roads; rivers and dams fill with mud.

5. Give methods of soil conservation.

Answer: Mechanical: **terracing** (flat steps cut into a slope, as on the Mandara Mountains), contour ploughing and ridges, stone lines and check dams. Biological: planting trees and **windbreaks**, cover crops, mulching, crop rotation and manure. Other: controlled grazing and stopping bush fires. **Strip cultivation** (strips of crops across the slope) also protects the soil.

C. THE ECOSYSTEM

6. Define an ecosystem and give its components.

Answer: An **ecosystem** is a natural area where living and non-living components interact. Components: abiotic (sunlight, water, air, soil, temperature); **producers** (green plants that make their own food from solar energy); consumers (animals); decomposers (fungi and bacteria).

7. How does energy flow in an ecosystem?

Answer: Energy comes from the Sun and is captured by producers, then passes to herbivores and carnivores. Only about **10 %** of the energy passes from one trophic level to the next; the rest is lost as heat. Example of a food chain: grass → zebra → lion; many chains linked make a food web.

D. CLIMATE DATA, SAVANNA AND HOT DESERT BIOMES

8. How do you work from a climate table?

Answer: Annual range of temperature = highest − lowest monthly temperature. Total rainfall = sum of the 12 monthly figures. The wettest month has the highest rainfall. Then compare with the biomes: rainforest (rain all year, range 2–3 °C), savanna (one wet and one dry season, 750–1,500 mm), hot desert (below 250 mm).

Practice: study the climate data of a station W in Africa (approximate values).

Month	J	F	M	A	M	J	J	A	S	O	N	D
Temp. (°C)	25	27	29	29	27	25	24	24	24	25	26	25
Rain (mm)	0	2	15	60	140	170	200	230	180	70	5	0

9. Calculate the annual range of temperature of station W, name its wettest month and name its biome with two reasons.

Answer: Range = 29 − 24 = **5 °C**. Wettest month: **August** (230 mm). Biome: **tropical wet and dry (savanna)**: hot all year (24 to 29 °C), one long rainy season (May to October) and a long dry season, with a total rainfall of about 1,070 mm.

10. Describe the savanna (location, vegetation, soils) and how its vegetation adapts to the dry season.

Answer: Between about 5° and 15° N and S, including northern Cameroon; tall grass with scattered trees; red ferruginous soils. Adaptations: the **baobab** stores water in its trunk; many trees shed their leaves; deep roots and thick bark against fire; grasses dry up and grow again with the first rains.

11. Describe the hot desert biome. Name its depositional features and ways to solve water scarcity.

Answer: Less than 250 mm of rain, very hot days and cold nights, sandy and stony soils, xerophytes (cacti, acacia). Wind deposition forms dunes: **barchans** (crescent-shaped) and seif; inselbergs, rock pedestals and yardangs are erosional features. Water scarcity: build **water reservoirs**, dig wells and boreholes, irrigate oasis farms.

E. CAMEROON: SITUATION AND RELIEF

12. Where is Cameroon located?

Answer: Between about **2° N and 13° N** and **8° E and 16° E**; in Central Africa, on the Gulf of Guinea. It covers about 475,000 km², is triangular in shape and has about 400 km of Atlantic coast.

13. Name the relief units of Cameroon.

Answer: Highlands: Western Highlands (Mount Cameroon, 4,095 m, the highest point), Adamawa Plateau (1,000–1,500 m), Mandara Mountains, South Cameroon Plateau. Lowlands: Coastal lowlands (Douala, Kribi), Benue lowland (Garoua), Chad Plain.

Quick quiz	Answer
Energy passed to the next trophic level	**About 10 %**
Organisms that make their own food	**Producers**
Crescent-shaped desert dune	**Barchan**
Cutting a slope into flat steps	**Terracing**

F. PRACTICE OF COMPETENCE

Situation

A widow farms a steep hill slope. After two years her yields fall because wind and heavy rain carry away the soil. She asks what is happening and what she should do.

14. What is the problem raised in the situation?

Answer: Her farm is losing soil through **soil erosion** (by rain and wind), so her yields are falling.

15. State two causes and propose two measures.

Answer: Causes: farming on a steep slope without protection; clearing the plant cover; heavy rain and strong winds; ploughing up and down the slope. Measures: terracing and contour ploughing; planting trees, windbreaks and cover crops; mulching and crop rotation.

Exam tip: *Questions begin with the verbs* ***State***, ***Name***, ***Identify***, ***Calculate****. Show your working for calculations and give a short reason for each answer.*